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Abstract:

This project presents MolGen, a diffusion-based Transformer framework for de novo molecular generation using SMILES representations. A discrete masking diffusion process is employed to progressively corrupt molecular sequences, enabling the model to learn structure-aware reconstruction under varying noise levels. The architecture integrates timestep-conditioned Transformer encoders to capture complex chemical dependencies and syntactic constraints. Generated molecules are evaluated using RDKit for validity and drug-likeness metrics, including QED and LogP. To ensure deployment efficiency, the trained model is exported to ONNX and optimized using TensorRT for accelerated inference on GPU platforms. The results demonstrate the feasibility of scalable, high-quality molecular generation for early-stage drug discovery.

Introduction:

The discovery and development of novel drug molecules is a complex, time-intensive, and costly process, often requiring extensive experimental screening and iterative optimization. Traditional computational approaches for molecular design rely on rule-based heuristics or exhaustive search strategies, which struggle to efficiently explore the vast chemical space, estimated to contain more than 10^{60} synthetically feasible compounds. This has motivated the adoption of data-driven generative models capable of proposing chemically valid and diverse molecular structures with minimal human intervention.

Recent advances in deep learning have significantly influenced molecular generation, with sequence-based models gaining particular prominence due to the availability of textual molecular representations such as Simplified Molecular Input Line Entry System (SMILES). By representing molecules as sequences of discrete tokens, SMILES enables the application of natural language processing techniques to chemical structure modeling. However, generating valid and chemically meaningful SMILES strings remains challenging, as minor syntactic errors can result in invalid molecules or chemically implausible structures. Consequently, generative models must effectively capture both the grammatical constraints of SMILES and the underlying chemical dependencies between atoms and functional groups.

Diffusion-based generative models have emerged as a powerful paradigm for data synthesis, demonstrating state-of-the-art performance in domains such as image generation and structured data modeling. Unlike autoregressive methods that generate sequences token by token, diffusion models learn to reverse a gradual noising process, allowing them to model complex global dependencies more effectively. While continuous diffusion processes are well-studied for image data, molecular sequences require discrete formulations that preserve syntactic correctness and chemical validity. In this work, we introduce **MolGen**, a discrete diffusion-based Transformer framework for de novo molecular generation using SMILES representations. MolGen employs a masking-based diffusion process in which molecular tokens are progressively replaced with a mask symbol according to a predefined noise schedule. A Transformer encoder conditioned on diffusion timesteps is trained to reconstruct the original molecular sequence from partially masked inputs, enabling the model to learn structure-aware denoising across multiple noise levels. By integrating positional embeddings and timestep embeddings, MolGen captures both sequential order and diffusion dynamics, facilitating robust molecular reconstruction.

Literature Review:

Generative modeling for molecular design has gained substantial attention as a promising approach to accelerate early-stage drug discovery. Early computational methods relied on rule-based systems and combinatorial chemistry, which were limited by handcrafted constraints and poor scalability. With the rise of machine learning, data-driven models have been increasingly adopted to learn molecular distributions directly from large chemical databases.

Initial deep learning approaches for molecular generation predominantly employed recurrent neural networks (RNNs) and long short-term memory (LSTM) architectures to model SMILES strings as sequential data. These models demonstrated the feasibility of learning chemical syntax and generating novel molecules; however, they often suffered from exposure bias and error accumulation due to their autoregressive nature. Small prediction errors early in the sequence could propagate, leading to invalid or chemically inconsistent molecules. Variational autoencoders (VAEs) were subsequently introduced to learn continuous latent representations of molecules, enabling interpolation and optimization in latent space. Despite their flexibility, VAE-based models frequently struggled with posterior collapse and limited control over molecular validity.

Graph-based generative models emerged as an alternative paradigm, representing molecules as graphs rather than sequences. These approaches explicitly modeled atomic connectivity and chemical bonds, improving structural validity. Nevertheless, graph-based models often required complex generation rules, were computationally expensive, and posed challenges in scaling to large molecular datasets. As a result, sequence-based methods using SMILES representations have remained attractive due to their simplicity, scalability, and compatibility with natural language processing techniques.

More recently, Transformer-based architectures have been applied to molecular generation, leveraging self-attention mechanisms to capture long-range dependencies within SMILES sequences. Transformers demonstrated improved performance over RNNs in modeling complex molecular structures and syntactic constraints. However, most Transformer-based models continued to rely on autoregressive decoding, inheriting limitations related to sequential dependency and limited global coherence during generation.

Diffusion models have emerged as a powerful class of generative models capable of learning complex data distributions by reversing a gradual noising process. In continuous domains such as images and audio, diffusion models have achieved state-of-the-art results due to their stable

training dynamics and strong generative capacity. Motivated by these successes, recent studies have explored diffusion-based approaches for molecular generation. Continuous diffusion methods applied to molecular graphs or latent representations have shown promise but often require intricate geometric constraints and expensive sampling procedures.

Discrete diffusion models offer a more natural formulation for sequence-based molecular representations. Masking-based diffusion approaches, inspired by masked language modeling and non-autoregressive generation, progressively corrupt discrete tokens and train models to reconstruct original sequences. Such methods have demonstrated advantages in capturing global structure, reducing error propagation, and enabling parallel generation. However, research on discrete diffusion models for molecular SMILES generation remains relatively limited, particularly with respect to efficient deployment and large-scale inference.

Furthermore, most existing molecular generative studies focus primarily on model accuracy and chemical validity, with comparatively little emphasis on deployment efficiency and real-world applicability. Model optimization techniques such as framework-agnostic export and hardware-aware acceleration are rarely integrated into molecular generation pipelines, despite their importance for practical adoption in pharmaceutical and industrial settings.

In light of these limitations, MolGen builds upon prior work in Transformer-based sequence modeling and discrete diffusion processes to provide a scalable and deployment-ready molecular generation framework. By combining timestep-conditioned Transformers with a masking-based diffusion strategy and incorporating efficient inference optimization, MolGen addresses key gaps in existing literature related to generation quality, global coherence, and practical usability.

Methodology:

MolGen is a discrete diffusion-based generative framework designed to synthesize novel molecular structures using SMILES representations. The methodology integrates sequence-based molecular encoding, a masking-driven diffusion process, and a timestep-conditioned Transformer architecture to learn the underlying distribution of chemically valid molecules. The complete pipeline consists of data preprocessing, discrete diffusion modeling, Transformer-based denoising, iterative molecule generation, chemical validation, and optimized deployment.

MolGen operates on the Simplified Molecular Input Line Entry System (SMILES), a widely adopted linear notation for representing molecular structures. Each molecule is expressed as a sequence of characters encoding atoms, bonds, ring closures, and branching information. Given the syntactic sensitivity of SMILES, even minor deviations can result in invalid molecular structures; therefore, precise sequence modeling is critical.

A character-level tokenization strategy is employed, where each unique SMILES character is mapped to a discrete token. Special tokens are introduced to support the diffusion process and sequence handling, including padding tokens for batch alignment, start-of-sequence and end-of-sequence markers, and a dedicated mask token used during diffusion. All tokenized sequences are padded or truncated to a fixed maximum length to enable efficient batch processing during training.

MolGen adopts a discrete diffusion formulation tailored for token-based molecular sequences. Unlike continuous diffusion models that add Gaussian noise, the forward diffusion process in MolGen progressively replaces tokens with a mask symbol according to a predefined noise schedule.

Given an original molecular sequence $x_0x_1\dots x_{L-1}$, the forward diffusion process generates a noised sequence $x_tx_{t+1}\dots x_{L-1}$ at timestep t by randomly masking tokens with a probability determined by a cosine-based schedule. At early timesteps, only a small fraction of tokens are masked, preserving most molecular information. As the timestep increases, the proportion of masked tokens grows, resulting in highly corrupted sequences that contain minimal structural information.

This progressive corruption forces the model to learn reconstruction under varying noise conditions, enabling robust generation capabilities.

The denoising component of MolGen is implemented using a Transformer encoder architecture. The Transformer is selected due to its ability to capture long-range dependencies and global contextual relationships within sequences, which are essential for modeling chemical structures such as ring closures and functional group interactions.

Each noised token sequence is mapped to a dense embedding space using token embeddings. Positional embeddings are added to preserve sequential ordering, ensuring that the relative positions of atoms and bonds are maintained. Additionally, a sinusoidal timestep embedding encodes the diffusion timestep t , allowing the model to condition its predictions on the noise level present in the input.

The embedded inputs are processed through multiple Transformer encoder layers consisting of multi-head self-attention and feed-forward sublayers. Self-attention enables the model to learn dependencies across the entire sequence simultaneously, facilitating coherent reconstruction of masked tokens based on global molecular context.

The final encoder outputs are projected onto the vocabulary space using a linear layer, producing a probability distribution over all possible tokens for each sequence position. The model is trained to predict the original token values corresponding to masked positions.

MolGen is trained using a masked cross-entropy loss function computed exclusively over masked tokens. This objective encourages the model to focus on reconstructing missing information rather than trivially copying unmasked tokens. For each training iteration, a random diffusion timestep is sampled, and the corresponding noised sequence is generated dynamically.

Optimization is performed using an adaptive optimizer with weight decay regularization. A cosine learning rate schedule is applied to stabilize training and improve convergence. Training is conducted on GPU hardware to accommodate large batch sizes and accelerate convergence.

During inference, MolGen generates novel molecules through an iterative reverse diffusion process. Generation begins from a fully masked sequence, representing maximum uncertainty. At each iteration, the model predicts token probabilities for all masked positions.

A confidence-based unmasking strategy is applied, wherein a subset of tokens with the highest prediction confidence is replaced at each step. This gradual refinement process continues until all tokens are resolved or a predefined number of steps is reached. Sampling techniques such as temperature scaling and top-k filtering are used to balance diversity and chemical validity in the generated molecules.

The generated SMILES strings are evaluated using cheminformatics tools to assess chemical validity and pharmaceutical relevance. Validity checks ensure that generated sequences correspond to chemically parsable molecules. Additional metrics such as quantitative estimates of drug-likeness and physicochemical properties are computed to assess the practical utility of the generated compounds.

To enable efficient real-world deployment, the trained MolGen model is exported to an interoperable intermediate format. Hardware-specific optimization techniques are applied to reduce inference latency and memory overhead. This optimization pipeline enables accelerated molecular generation on GPU platforms, demonstrating MolGen's suitability for large-scale and production-oriented environments.



Figure 1.1: MolGen Block Diagram

The block diagram illustrates the MolGen architecture, the SMILES dataset is first tokenized and then goes through a diffused discrete masking process, a transformer denoiser is used to remove noise and it comprises of encoder and decoder architecture, the reverse diffusion generation is used to generate new molecules and then further chemically validated before deployment.

Results:

The performance of the proposed MolGen framework was evaluated using a combination of validity, uniqueness, novelty, and physicochemical property analysis on a set of generated molecular structures. A total of 500 molecular samples were generated through the reverse diffusion process and subsequently assessed using cheminformatics-based evaluation metrics.

Out of the 500 generated SMILES strings, 340 molecules were found to be chemically valid, resulting in a validity rate of **68.0%**. This indicates that a substantial majority of the generated sequences conform to syntactic and chemical constraints required for successful molecular parsing. Among the valid molecules, all 340 were unique, corresponding to a **100% uniqueness score**, demonstrating that MolGen avoids mode collapse and does not redundantly reproduce identical molecular structures during generation.

Novelty was evaluated by comparing generated molecules against the training dataset. Of the 340 unique valid molecules, 312 were not present in the original dataset, yielding a **novelty rate of 91.8%**. This highlights MolGen's ability to explore new regions of chemical space while maintaining chemical plausibility, an essential requirement for de novo drug discovery applications.

In addition to structural metrics, the generated molecules were evaluated for physicochemical properties relevant to drug-likeness. The quantitative estimate of drug-likeness (QED) across the valid molecules exhibited a mean value of **0.743 with a standard deviation of 0.140**. The distribution of QED scores was skewed towards higher values, indicating that a large proportion of generated molecules possess favorable drug-like characteristics according to established pharmaceutical heuristics.

Lipophilicity, measured using LogP, yielded a mean value of **2.577 with a standard deviation of 1.453**. The observed LogP distribution primarily lies within the range commonly associated with acceptable bioavailability and membrane permeability, suggesting that MolGen-generated molecules are chemically balanced with respect to hydrophobic and hydrophilic properties.

Representative molecular structures generated by MolGen further illustrate the diversity and complexity of the synthesized compounds. These molecules exhibit varied ring systems, functional groups, and structural motifs, reflecting the model's ability to capture non-trivial chemical patterns. Collectively, these results demonstrate that MolGen successfully generates diverse, novel, and drug-like molecules with a high degree of chemical validity.

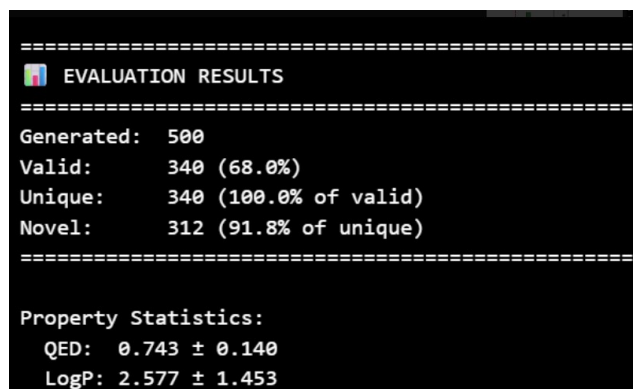


Figure 1.2 : Results & Statistics of MolGen

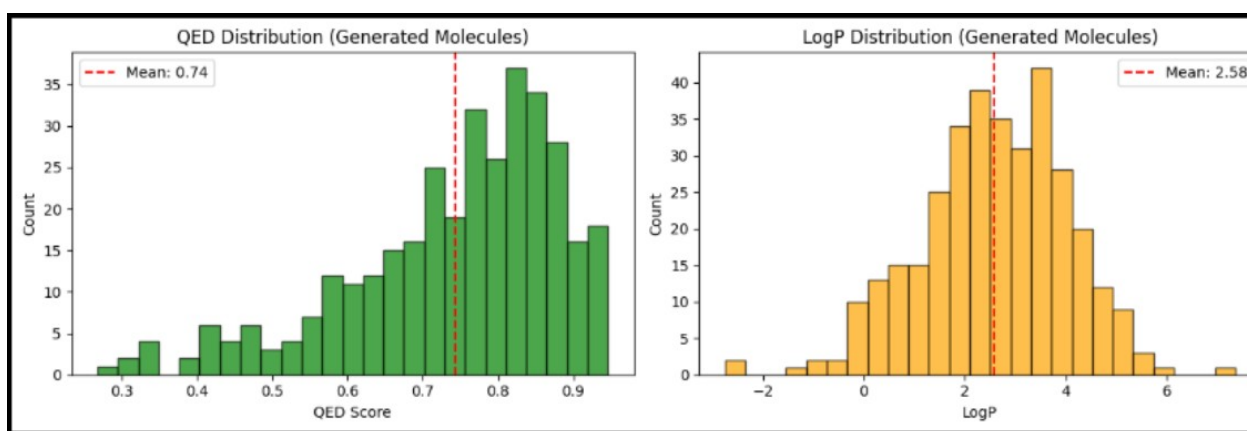


Figure 1.3 : QED and LogP Distribution in MolGen

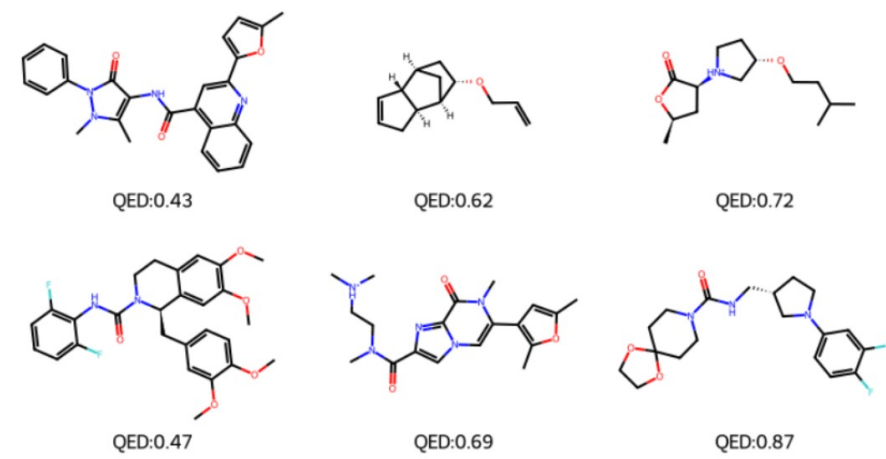


Figure 1.4 Newly Generated Molecules

Discussion:

The experimental results demonstrate that MolGen effectively combines discrete diffusion modeling with Transformer-based sequence learning to generate chemically meaningful molecular structures. The achieved validity rate of 68.0% is competitive for sequence-based generative models operating directly on SMILES representations, particularly given the non-autoregressive nature of the generation process. The presence of invalid molecules can be attributed primarily to the inherent syntactic sensitivity of SMILES and the difficulty of enforcing strict grammar constraints in fully generative settings.

The observed 100% uniqueness among valid molecules highlights a key strength of the diffusion-based generation paradigm. Unlike autoregressive models that may repeatedly converge to high-probability modes, MolGen's iterative denoising mechanism promotes global diversity by refining multiple tokens simultaneously. This property is especially advantageous for early-stage drug discovery, where exploration of diverse chemical candidates is critical.

High novelty performance further emphasizes MolGen's generative capacity. With over 91% of unique molecules being absent from the training data, the model demonstrates an ability to generalize beyond memorization and synthesize genuinely new chemical structures. This suggests that MolGen learns a meaningful underlying distribution of molecular patterns rather than simply reproducing known compounds.

The physicochemical property analysis provides additional insight into the quality of generated molecules. The mean QED score of 0.743 indicates that many generated compounds satisfy multiple drug-likeness criteria simultaneously, such as molecular weight, lipophilicity, and structural complexity. The LogP distribution centered around moderate values suggests a favorable balance between solubility and permeability, which is a key consideration in pharmacokinetics.

Despite these strengths, certain limitations remain. The validity rate, while reasonable, indicates room for improvement through enhanced syntax-aware constraints or hybrid graph-sequence modeling. Additionally, while QED and LogP provide useful high-level indicators, future work could incorporate target-specific objectives, toxicity prediction, or multi-objective optimization to further enhance applicability.

Overall, MolGen demonstrates that discrete diffusion Transformers constitute a promising and scalable approach for molecular generation.

Conclusion:

This work presented **MolGen**, a discrete diffusion-based Transformer framework for de novo molecular generation using SMILES representations. By formulating molecular generation as a masked token denoising task conditioned on diffusion timesteps, MolGen effectively bridges advances in sequence modeling and generative diffusion processes. The proposed methodology enables global context modeling, mitigates error propagation commonly observed in autoregressive approaches, and supports parallel refinement of molecular structures.

Experimental evaluation demonstrates that MolGen is capable of generating chemically valid, diverse, and novel molecules. A validity rate of 68.0% indicates that the model successfully learns syntactic and chemical constraints inherent to SMILES, while achieving complete uniqueness among valid molecules highlights its robustness against mode collapse. Furthermore, a novelty rate exceeding 90% confirms MolGen's ability to explore previously unseen regions of chemical space rather than reproducing training samples.

Physicochemical analysis further validates the quality of the generated molecules. The observed QED distribution indicates strong drug-likeness across a majority of valid samples, while the LogP values suggest favorable balance between solubility and permeability. These results collectively demonstrate that MolGen produces molecules that are not only structurally diverse but also aligned with pharmaceutical design principles.

An additional strength of this work lies in its emphasis on deployment readiness. By exporting the trained model to an interoperable representation and applying hardware-specific optimization techniques, MolGen enables accelerated inference suitable for large-scale molecular screening pipelines. This practical focus distinguishes the proposed framework from many prior generative chemistry models that remain confined to experimental settings.

While the results are encouraging, further improvements are possible. Enhancing chemical validity through syntax-aware constraints, incorporating multi-objective optimization, and extending the framework to conditional or target-specific molecular generation represent promising directions for future work. Overall, MolGen establishes discrete diffusion Transformers as a scalable, efficient, and effective solution for molecular generation, contributing meaningfully to the advancement of AI-driven drug discovery.

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